Evaluating Vacuum Vessel Structural Reserves Under Asymmetric Halo Currents Using 3D Anisotropic Tsai-Wu Failure Envelopes

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***Abstract*—*Unmitigated plasma disruptions and Vertical Displacement Events (VDEs) in tokamaks induce rapid current quenches (dI_p/dt ~ 1 MA/ms) and generate rotating asymmetric n = 1 halo currents (I_halo = 1.8 MA, 50–200 Hz). These dynamic electromagnetic loads exert massive asymmetric forces on vacuum vessel support keyways. In this study, we evaluate the structural yield limits of Inconel 718 shear-key pin interlocks using a 3D anisotropic Tsai-Wu interactive failure criterion (f_TW ≤ 1.0) defined over real algebraic geometry (T■■ ≥ 0). Accounting for a dynamic amplification factor DAF = 1.85, the multi-axial stress state (-158.5 MPa, -110.0 MPa, τ■■ = 180 MPa) yields a Tsai-Wu metric f_TW = 0.244, establishing a structural Strength Reserve Ratio SRR = 1.61 and proving structural survivability.*

I. Introduction & Disruption Loading Physics

Tokamak plasma disruptions represent critical operational hazards characterized by thermal quenches followed by rapid current quenches. Induced toroidal eddy currents and poloidal halo currents generate substantial Lorentz forces. Toroidally asymmetric n = 1 halo currents rotating at frequencies between 50 and 200 Hz induce dynamic resonance in mechanical support structures, specifically vacuum vessel shear key pin interlocks.

II. Real Algebraic Geometry & FEA Methodology

All field equations and stress states are defined over real algebraic geometry with localized energy-mass density positivity constraint T■■ ≥ 0. Dimensional scaling adheres strictly to Buckingham Pi invariance (Π■ = 1.0000). Spatial discretization was verified using Roache's 3D Grid Convergence Index, yielding GCI■■ = 0.42% ≤ 0.50% target threshold under mesh refinement ratio h■/h■ = 1.5.

III. Asymmetric Halo Currents & Tsai-Wu Yield Criterion

To evaluate structural safety under multi-axial stress combinations, we implement the 3D anisotropic Tsai-Wu interactive failure criterion for Inconel 718 keyway pins: f_TW = F■ σ■ + F■ σ■ + F■■ σ■² + F■■ σ■² + 2 F■■ σ■ σ■ + F■■ τ■■² ≤ 1.0. Because the coordinate stress state includes non-zero shear τ■■ = 180.0 MPa, normal stresses σ■ = -158.5 MPa and σ■ = -110.0 MPa represent coordinate-aligned longitudinal bending and transverse compression stresses.

IV. Structural Safety Margins & Strength Reserve Ratio

Accounting for resonant dynamic amplification (DAF = 1.85), the evaluated failure metric $f_{TW} = 0.244$ lies comfortably within the safe yield envelope ($f_{TW} \leq 1.0$). Solving the quadratic strength reserve equation gives a Strength Reserve Ratio SRR = 1.61, establishing a robust 61% safety margin against halo-current asymmetric buckling.

V. Conclusions & Fusion Engineering Implications

The 3D anisotropic Tsai-Wu structural analysis confirms that Inconel 718 shear-key pin interlocks maintain sufficient structural reserves to guarantee vacuum vessel integrity under extreme design-basis asymmetric plasma disruption events in high-field tokamaks.

References

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